

```

Process p = Runtime.getRuntime().exec(s);
BufferedReader stdInput = new BufferedReader(new
InputStreamReader(p.getInputStream()));
while ((s = stdInput.readLine()) != null)
{
System.out.println(s);
}

```

Let's first create a string to store our command line, concatenating variables for the username, password and message in the necessary order. Next, execute the command using the `getRuntime().exec()` function. The output of the command can be accessed via the `getInputStream()` method of the process variable.

You can do a search using the standard Java XML parsing methods, passing the search URL as the XML source. Here is the code for this:

```

DocumentBuilder builder = DocumentBuilderFactory.newInstance().
newDocumentBuilder();
URL u = new URL("http://search.twitter.com/search.atom?q=%23worldcup");
Document doc = builder.parse(u.openStream());
String vTemp=null;
String vName=null;
NodeList nodes = doc.getElementsByTagName("entry");
NodeList temp = doc.getElementsByTagName("author");
for(int i=0;i<nodes.getLength();i++)
{
    Element element = (Element)nodes.item(i);
    Element element_t = (Element)temp.item(i);
    vName=getElementValue(element_t,"name").toString();
    vTemp+=vName;
    vTemp+=getElementValue(element,"title").toString();
}

```

Step 3: Desktop client

Now we have a Java program to post tweets; let's add a GUI, using the NetBeans IDE, and make it more usable and professional-looking. First, create a new 'Java Desktop Application' using the NetBeans new project wizard. You will see a blank window/palette to design your form. Insert a tabbed pane to the blank window. Create one more tab, and add two panels in each of the two tabs. In one of the panels, add three text boxes, with the corresponding labels, and the push-button 'Tweet'. In another panel, add a text box with a label, a text area and the push button 'Search'. After completing the design, the GUI should look something like what's shown in Figure 1.

Once the UI is done, go to the events of the Tweet and Search buttons. For the mouse-click event, add the corresponding code from Step 2. Replace the variables for the *username*, password and message with the string values from the input fields in the form. For example, replace the username variable with `jUsername.getText()` (where `jUsername` is the name of the text box for the username).

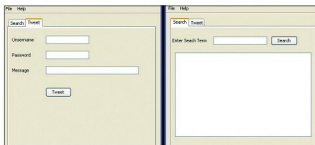


Figure 1: Completed GUI design

The final code for sending a tweet will look something like what's given below:

```

private void jButton2MouseClicked(java.awt.event.MouseEvent evt)
{
    String message = jMessage.getText().toString().replaceAll(" ", "%20");
    String s = "curl -basic -user "+jUsername.getText()+"-"+jPassword.getText()+"
--data status="+message+" "+http://twitter.com/statuses/update.xml";

    Process p = Runtime.getRuntime().exec(s);
    BufferedReader stdInput = new BufferedReader(new
InputStreamReader(p.getInputStream()));
    while ((s = stdInput.readLine()) != null)
    {
        System.out.println(s);
    }
    System.exit(0);
}

```



Tip: The `replaceAll` function replaces all spaces with % followed by the ASCII hexadecimal equivalent code, since spaces cannot be sent in the HTTP request.

To retrieve the search items, the string `vTemp` in the search code in Step 2 can be passed to the text area in the search tab, using:

```
jResult.setText(vTemp);
```

Thus, using the simple Twitter API and an easy-to-use Java IDE, you can develop a Twitter client in just a few minutes. With some more tweaking and additions to the code, you can easily add other features such as re-tweeting, direct messages, follow/unfollow, and URL shortening. You can download the source code of the tutorial from <http://code.google.com/p/twidesk/>. 

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